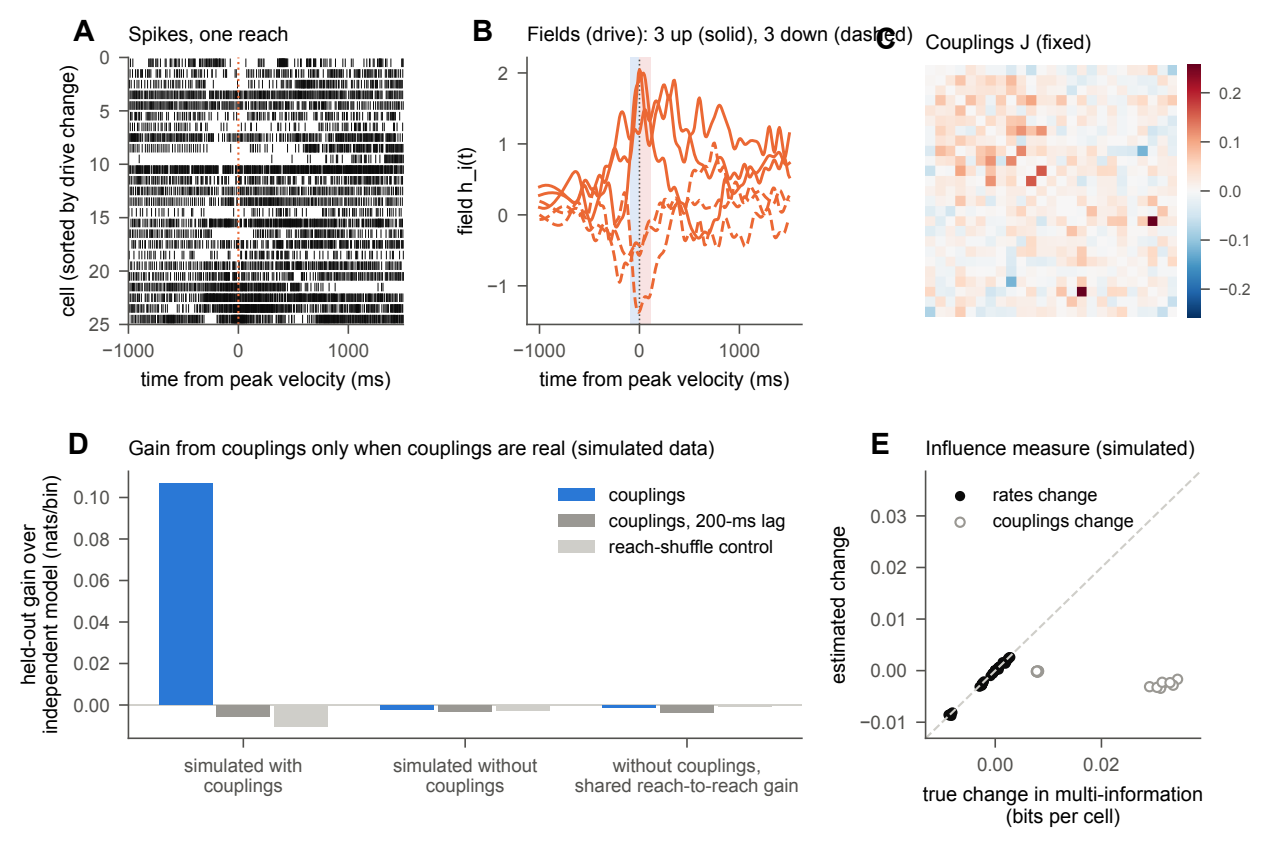
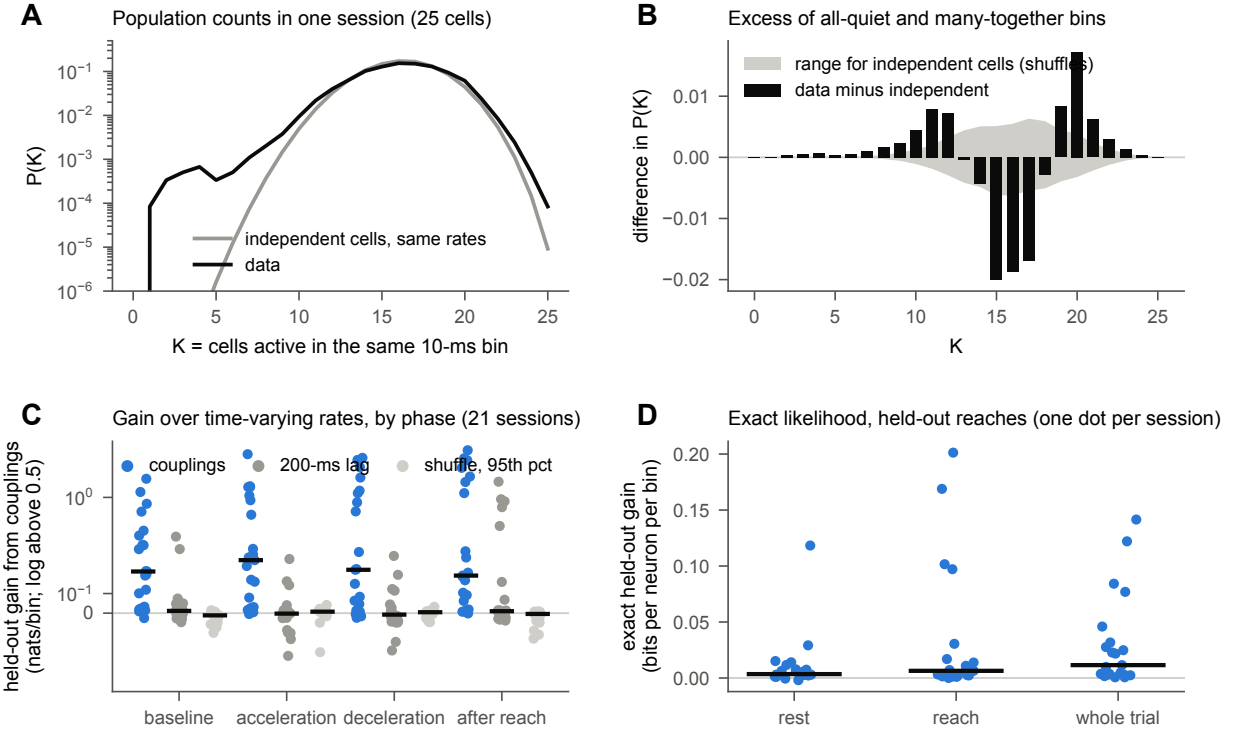
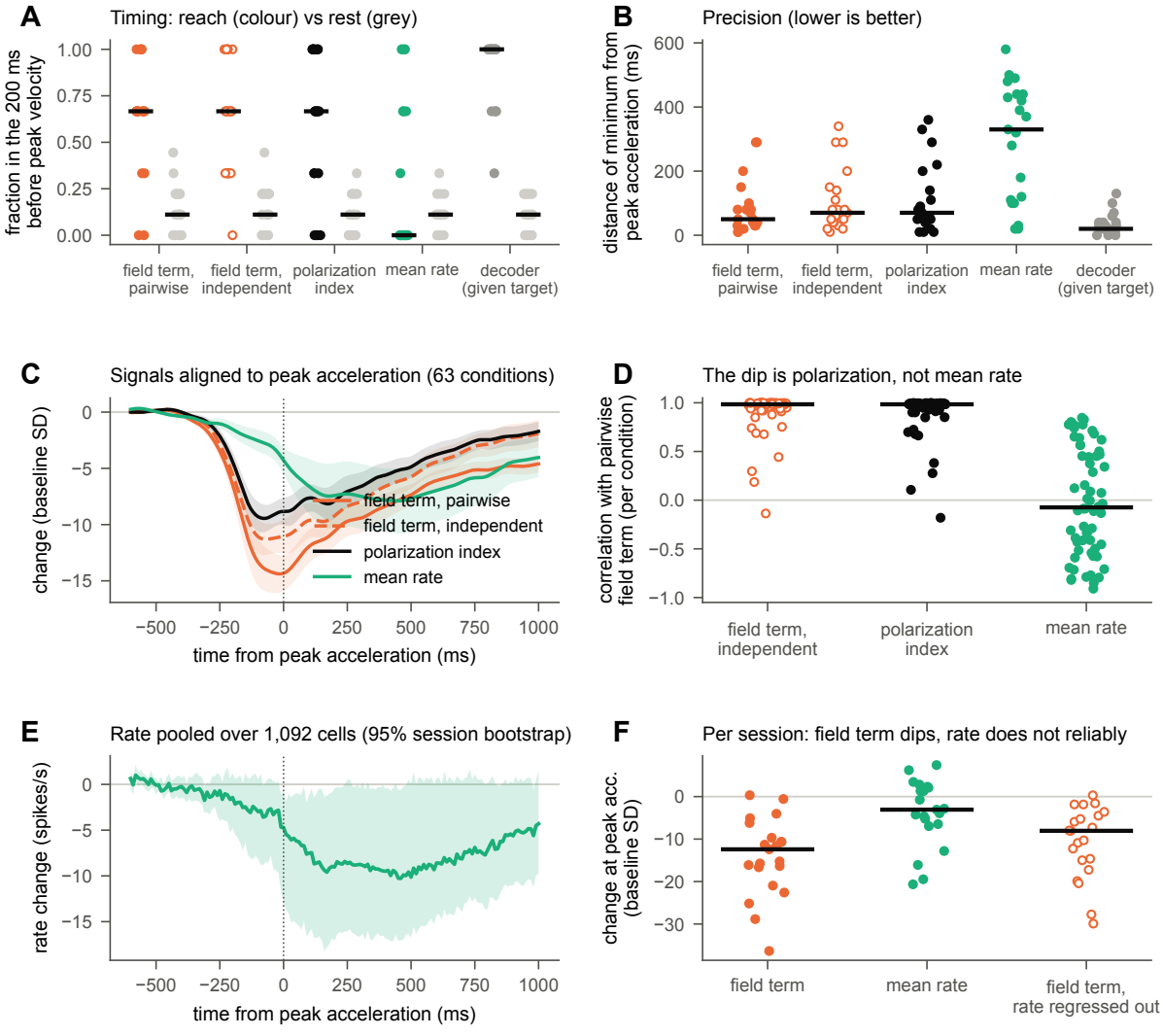
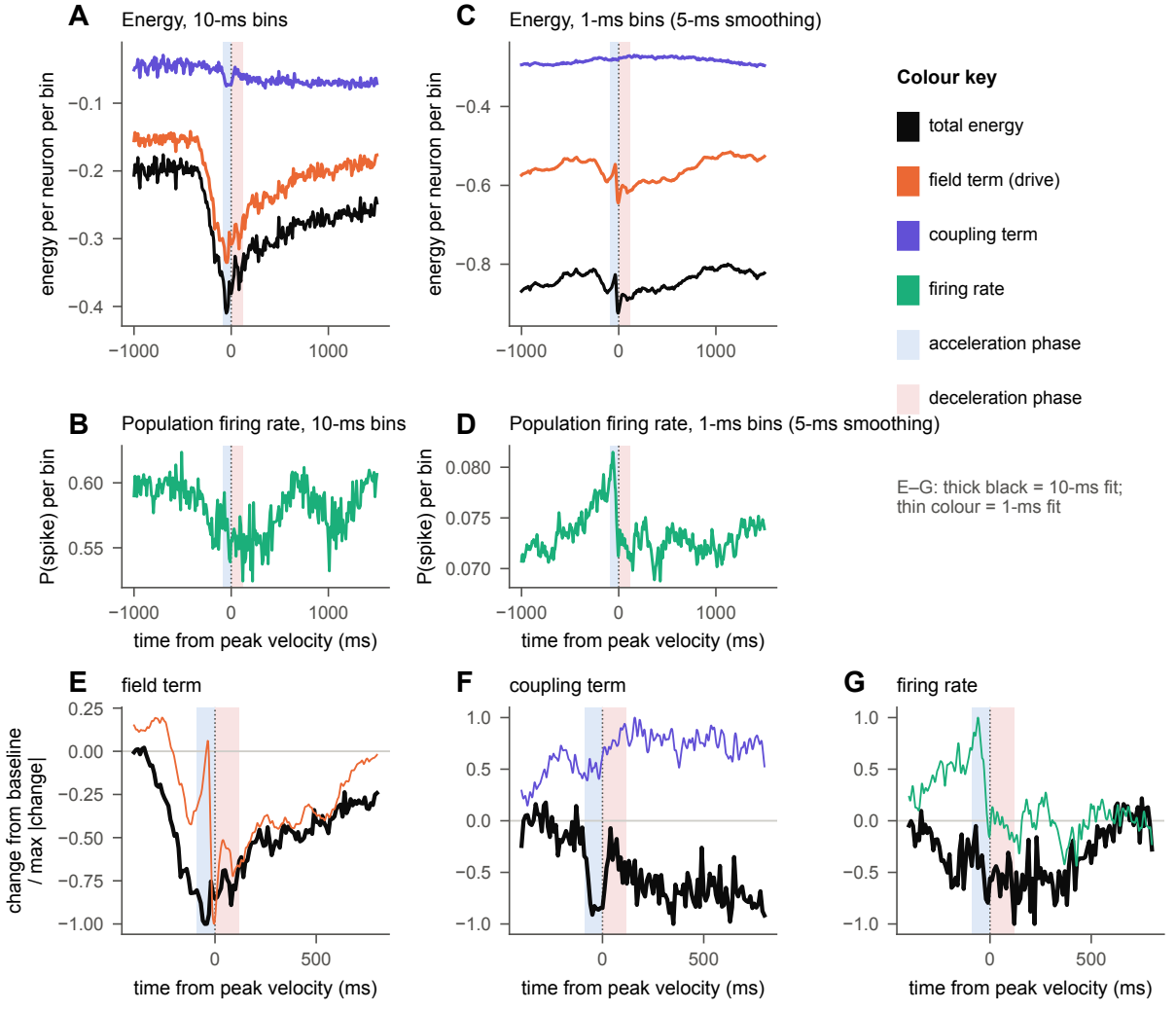
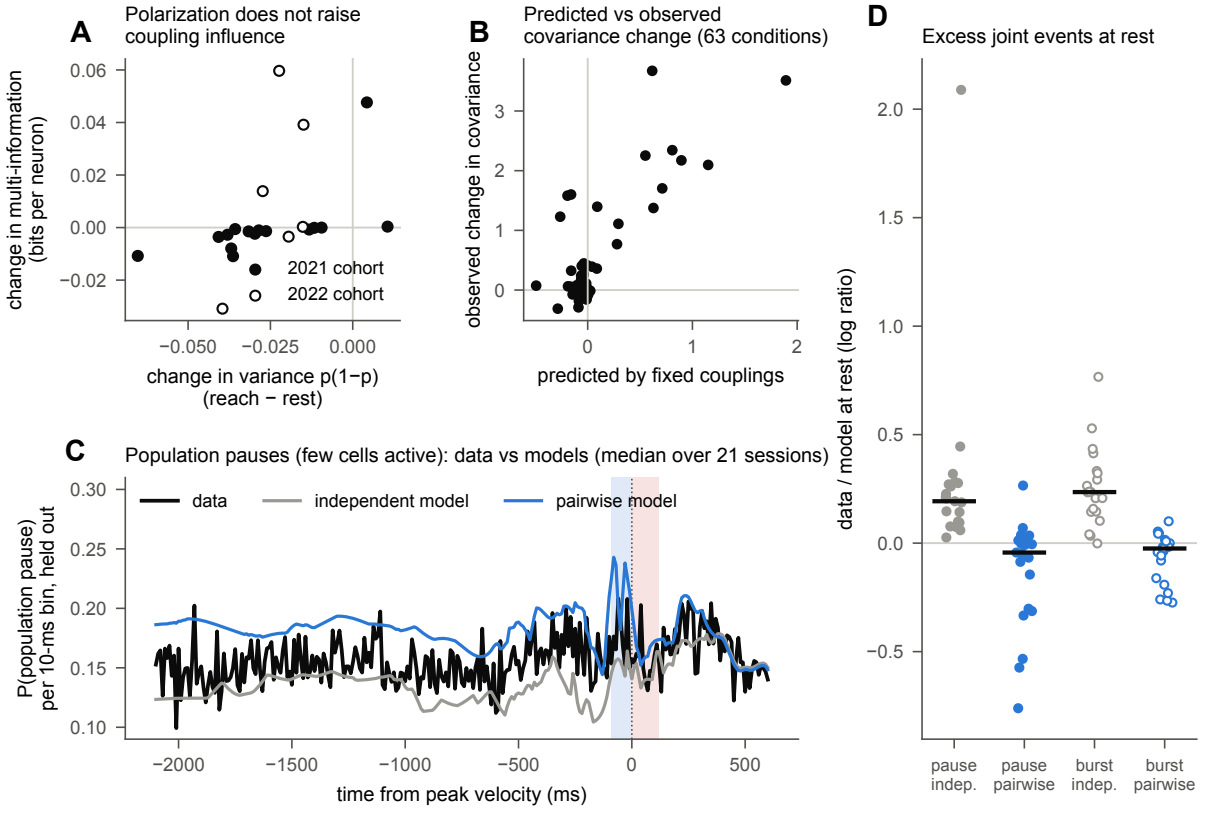
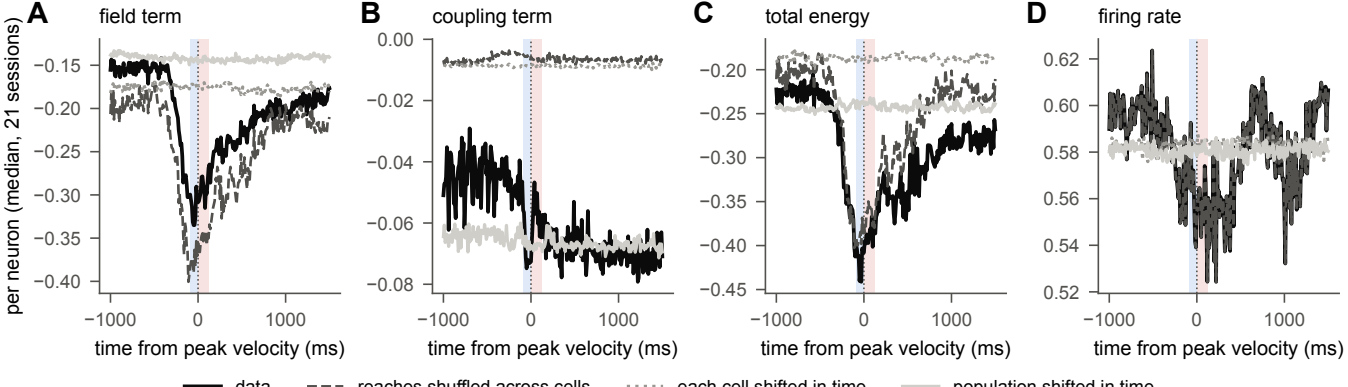

A
Spikes, one reach
cell (sorted by drive change)
0
5
10
15
20
25
−1000
0
1000
time from peak velocity (ms)
B
Fields (drive): 3 up (solid), 3 down (dashed)
field h_i(t)
2
1
0
−1
−1000
0
1000
time from peak velocity (ms)
C
Couplings J (fixed)
0.2
0.1
0.0
−0.1
−0.2
D
Gain from couplings only when couplings are real (simulated data)
held-out gain over independent model (nats/bin)
0.10
0.08
0.06
0.04
0.02
0.00
couplings
couplings, 200-ms lag
reach-shuffle control
simulated with couplings
simulated without couplings
without couplings, shared reach-to-reach gain
E
Influence measure (simulated)
rates change
couplings change
estimated change
0.03
0.02
0.01
0.00
−0.01
0.00
0.02
true change in multi-information (bits per cell)

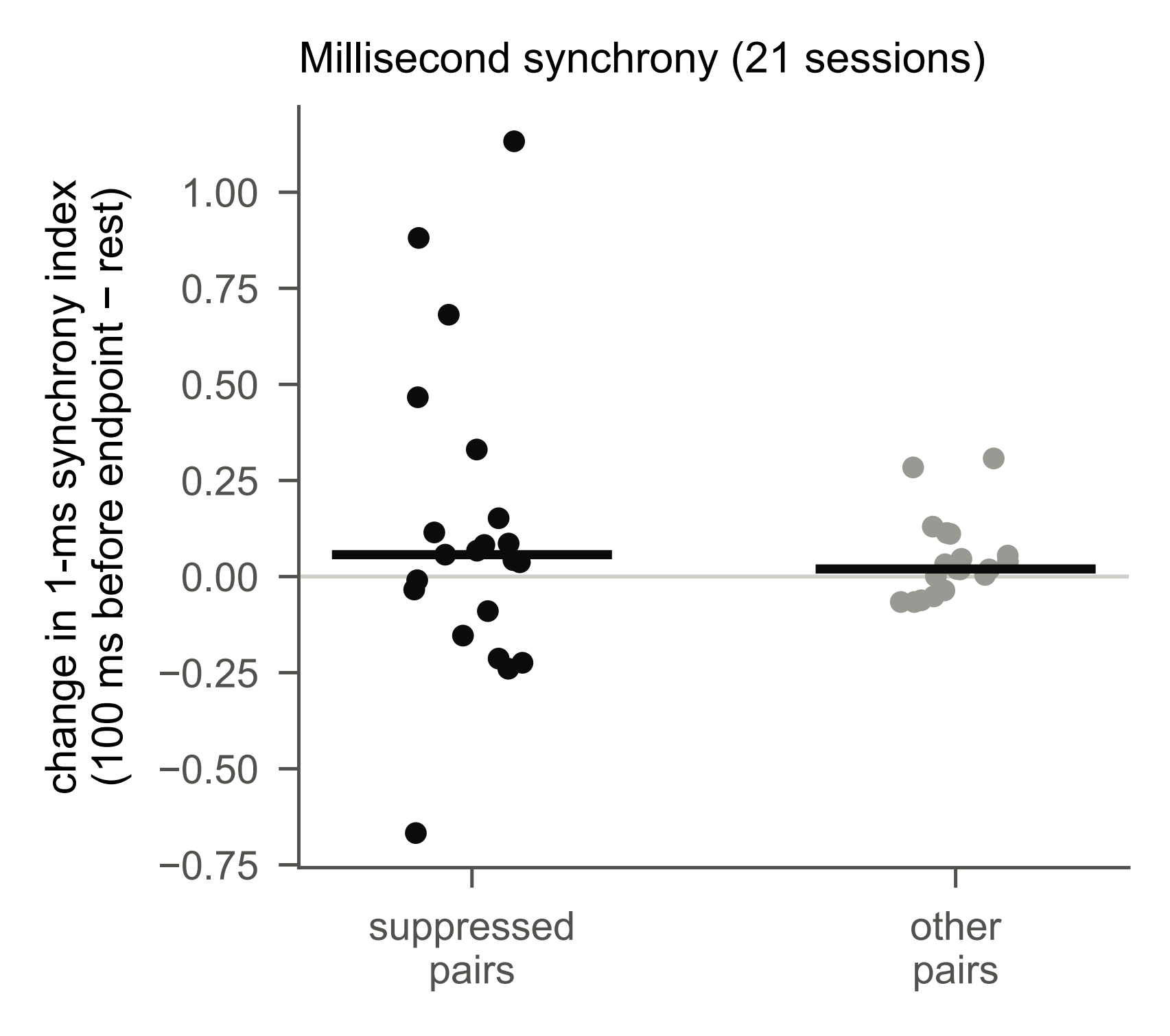

A
Population counts in one session (25 cells)
P(K)
independent cells, same rates
data
K = cells active in the same 10-ms bin
B
Excess of all-quiet and many-together bins
range for independent cells (shuffles)
data minus independent
difference in P(K)
K
C
Gain over time-varying rates, by phase (21 sessions)
held-out gain from couplings (nats/bin; log above 0.5)
couplings
200-ms lag
shuffle, 95th pct
baseline
acceleration
deceleration
after reach
D
Exact likelihood, held-out reaches (one dot per session)
exact held-out gain (bits per neuron per bin)
rest
reach
whole trial


A
Timing: reach (colour) vs rest (grey)
fraction in the 200 ms before peak velocity
1.00
0.75
0.50
0.25
0.00
field term, pairwise
field term, independent
polarization index
mean rate
decoder (given target)
B
Precision (lower is better)
distance of minimum from peak acceleration (ms)
600
400
200
0
field term, pairwise
field term, independent
polarization index
mean rate
decoder (given target)
C
Signals aligned to peak acceleration (63 conditions)
change (baseline SD)
0
−5
−10
−15
field term, pairwise
field term, independent
polarization index
mean rate
−500
−250
0
250
500
750
1000
time from peak acceleration (ms)
D
The dip is polarization, not mean rate
correlation with pairwise field term (per condition)
1.0
0.5
0.0
−0.5
−1.0
field term, independent
polarization index
mean rate
E
Rate pooled over 1,092 cells (95% session bootstrap)
rate change (spikes/s)
0
−5
−10
−15
−500
−250
0
250
500
750
1000
time from peak acceleration (ms)
F
Per session: field term dips, rate does not reliably
change at peak acc. (baseline SD)
0
−10
−20
−30
field term
mean rate
field term, rate regressed out


A
Energy, 10-ms bins
energy per neuron per bin
−0.1
−0.2
−0.3
−0.4
−1000
0
1000
C
Energy, 1-ms bins (5-ms smoothing)
energy per neuron per bin
−0.4
−0.6
−0.8
−1000
0
1000
Colour key
total energy
field term (drive)
coupling term
firing rate
acceleration phase
deceleration phase
B
Population firing rate, 10-ms bins
P(spike) per bin
0.60
0.55
−1000
0
1000
time from peak velocity (ms)
D
Population firing rate, 1-ms bins (5-ms smoothing)
P(spike) per bin
0.080
0.075
0.070
−1000
0
1000
time from peak velocity (ms)
E–G: thick black = 10-ms fit;
thin colour = 1-ms fit
E
field term
change from baseline
/ max |change|
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
500
time from peak velocity (ms)
F
coupling term
1.0
0.5
0.0
−0.5
−1.0
0
500
time from peak velocity (ms)
G
firing rate
1.0
0.5
0.0
−0.5
−1.0
0
500
time from peak velocity (ms)


A
Polarization does not raise coupling influence
change in multi-information (bits per neuron)
0.06
0.04
0.02
0.00
−0.02
2021 cohort
2022 cohort
−0.050
−0.025
0.000
change in variance p(1−p) (reach − rest)
B
Predicted vs observed covariance change (63 conditions)
observed change in covariance
3
2
1
0
0
1
2
predicted by fixed couplings
C
Population pauses (few cells active): data vs models (median over 21 sessions)
P(population pause) per 10-ms bin, held out
0.30
0.25
0.20
0.15
0.10
data
independent model
pairwise model
−2000
−1500
−1000
−500
0
500
time from peak velocity (ms)
D
Excess joint events at rest
data / model at rest (log ratio)
2.0
1.5
1.0
0.5
0.0
−0.5
pause indep.
pause pairwise
burst indep.
burst pairwise


A
field term
B
coupling term
C
total energy
D
firing rate
per neuron (median, 21 sessions)
−0.15
−0.20
−0.25
−0.30
−0.35
−0.40
0.00
−0.02
−0.04
−0.06
−0.08
−0.20
−0.25
−0.30
−0.35
−0.40
−0.45
0.62
0.60
0.58
0.56
0.54
0.52
−1000
0
1000
time from peak velocity (ms)
data
reaches shuffled across cells
each cell shifted in time
population shifted in time


Millisecond synchrony (21 sessions)
change in 1-ms synchrony index
(100 ms before endpoint – rest)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
suppressed
pairs
other
pairs